



Using the cosine ratio

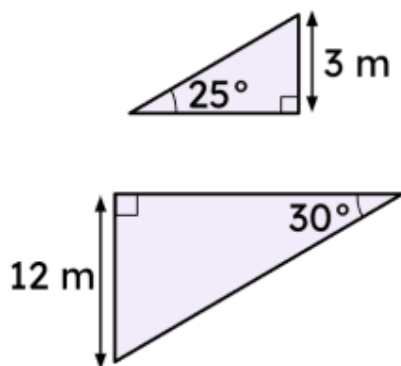
1 In a right triangle, if the hypotenuse is 75 cm and a second side is 60 cm, what is the length of the third side? (Use a calculator to help you.) Tick **1** correct answer

☐ 50 cm

☐ 45 cm

☐ 54 cm

2 Which of the following statements is true for these triangles? Tick **1** correct answer



☐ The triangles are similar.

☐ The triangles are not similar.

☐ It is not possible to know whether the triangles are similar or not.

3 In a right triangle, if the shortest sides are 7.5 cm and 18 cm, what is the length of the hypotenuse? (Use a calculator to help you.) Tick **1** correct answer

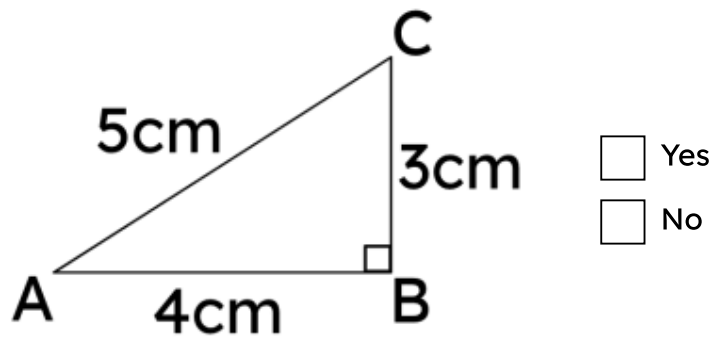
☐ 19.5 cm

☐ 18.5 cm

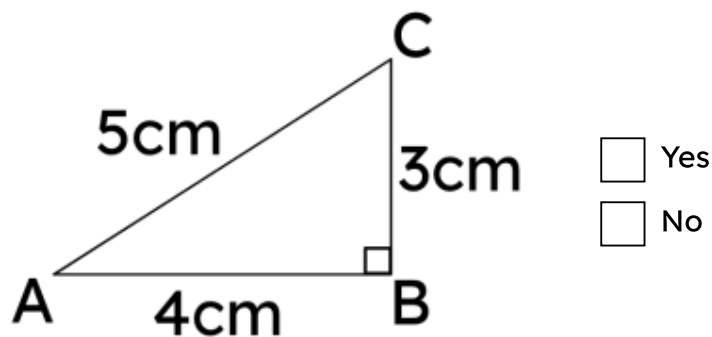
☐ 20.5 cm



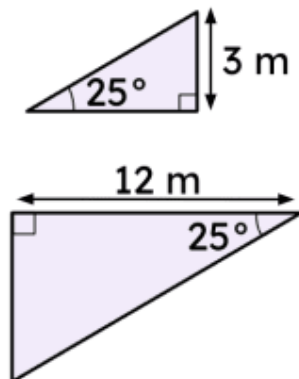
- 4 Would a triangle ABC with sides $AB = 16\text{cm}$, $BC = 10\text{cm}$, $AC = 21\text{cm}$ be similar to the one shown in the diagram? Tick **1** correct answer



- 5 Would a triangle ABC with sides $AB = 12\text{ cm}$, $BC = 9\text{ cm}$, $AC = 15\text{ cm}$ be similar to the one shown in the diagram? Tick **1** correct answer



- 6 For this pair of triangles, can you determine whether they are similar without using side lengths? Tick **1** correct answer



- ☐ Yes, because their three angles correspond.
- ☐ No because you only know two angles.
- ☐ No, because you always need a side and two angles.

